

PITCH DECK

Hackathon Presentation

Simple English Edition

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DATA CLEAN AGENT

Cleaning Messy Data in Seconds with Safe, Trustworthy AI

[Judge Pitch Guide & Live Demo Cheat Sheet](#)

Dirty Data is a Nightmare

Why businesses struggle to use raw data and why AI cleaning tools are often rejected.

- **Data scientists waste 80% of their time**

They spend hours manually fixing typos, handling missing values, and clipping outliers instead of building models.

- **Traditional tools are a Black Box**

Standard AI cleaning tools just return the cleaned file. You don't know what they changed, what they deleted, or if they altered critical details.

- **No Trust = No Adoption**

If companies cannot see the exact mathematical steps the AI took, they cannot verify it and will not deploy it to production.

Safe and Smart AI

How DataCleanAgent bridges the gap between intelligence, safety, and transparency.

- **Step 1: Multi-Dimensional Profiling**

Checks your dataset against 6 core quality metrics (Completeness, Uniqueness, Validity, Consistency, Accuracy, Integrity).

- **Step 2: AI Planning (ReAct loop)**

The AI inspects the errors, plans the correct cleaning action, and executes it step-by-step.

- **Step 3: Deterministic Action Registry**

The AI cannot run arbitrary code on your PC. It only selects from pre-built, safe python actions (like mode-imputation, outliers clipping, etc.).

- **Step 4: Safety Verifier**

Ensures the AI doesn't accidentally change the schema or drop too many rows. If a step fails, the AI self-heals and corrects it.

Premium Extras that Wow

Features we added to prove this is an enterprise-ready product, not just a toy.

- **Unified Data Health Score Gauge**

A live Plotly dial that visualizes the dataset health rating from 0-100%. Watch the needle move from 'Before' to 'After' during the cleaning process!

- **Download Reproducible Python Script**

Generates the exact, standalone pandas code the AI applied. Data engineers can download and run this script locally in their systems.

- **Session Rollback Support**

Interactively reverse any cleaning step. If the user doesn't like a change, one click rolls back the data and the AI's chat memory.

Winning Over the Judges

A step-by-step walkthrough to showcase the platform's power in a live presentation.

- **1. Ingestion: Upload the test dataset**

Choose 'Upload CSV File' and upload 'random_dirty_sales.csv' from your local data folder to show a raw, dirty file.

- **2. Run the Loop: AI in action**

Click 'Run Cleaning Loop'. Point to the terminal output showing the AI reasoning, self-healing, and applying sanitization steps.

- **3. Dashboard Comparison: Tab 3**

Click 'Tab 3: Compare'. Show the health gauge score jump (e.g. from 62% to 98%) and explain the radar chart overlay.

- **4. Code Export: Transparency**

Scroll down to 'Audit & Export' and download the Python Script. Show the judges the clean pandas code generated on the fly.

Behind the Scenes

The technical foundation that ensures the application is secure, fast, and bulletproof.

- **Tech Stack**

Streamlit for front-end, Google Gemini 2.5 Flash for reasoning, SQLite for sessions, and Plotly for visualizations.

- **100% Secure Design**

Immune to code injection because the LLM is restricted to calling registry arguments. No raw python string executions are allowed.

- **Enterprise Stability**

A comprehensive testing suite with 48/48 automated unit tests fully passing. Robust error handling ensures the app never crashes.